



A secure blockchain-based group key agreement protocol for IoT

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Abstract

Since most Internet of things (IoT) devices are energy-limited, increasingly more manufacturers have chosen to develop IoT applications based on group communication. For these applications, a secure and effective authenticated group secret key is fundamental to whole group communication. A large number of related protocols have been proposed to address this security problem. However, most of them have a centralized architecture, which is vulnerable to a single point of failure. Blockchain technology has the potential to address the challenges mentioned above because of its distributed, secure, and private property. Thus, this paper proposes a blockchain-based authenticated group key agreement protocol for IoT. The proposed protocol introduces a new entity called the device manager, who acts as an intermediary to connect IoT devices with blockchain networks. Security analysis demonstrates the robustness of the proposed protocol to various kinds of attacks. Besides, the simulation results further show that the time costs of protocol operations are reasonable and suitable for IoT environments.

Keywords Internet of things · Blockchain · Group communication · Security · Private key

1 Introduction

With the advent of internet of things (IoT) technologies, a large number of intelligent devices have been developed and integrated into daily life [1–3]. IoT technologies have achieved unprecedented success both concerning scalability and applicable scenarios. This emerging technology provides innovative applications such as wireless sensor networks [4], health care [5], smart grid [6], and VANET [7]. Because of the constraints that most IoT devices operate under (limited energy or low storage),

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increasingly more device manufacturers prefer to launch group-communication-based applications (e.g., smart-home and health-care systems) to increase profits. However, with the increasing numbers of devices and the complexity of the communication among them, IoT applications, especially those based on group communication, have gradually become the main target of hacking attacks.

To ensure communication security, a common practice is to set up a group key for the whole group among devices [8–10]. This ensures that only devices with the group key can access data within the group. As of today, several group key agreement protocols have been proposed for IoT. Most of these protocols employ a centralized architecture [11–13], in which a central key server is responsible for securely and efficiently distributing the group key to group members. However, there is a risk that private data stored in the central key server may be attacked. Given the distributed nature of the IoT network, a distributed (or decentralized) model seems more suitable for IoT environments. Several distributed group key agreement protocols for IoT have been introduced [14, 15]. These protocols involve all nodes working together to generate and manage group key(s). As there is no central authority that acts as a trust key server, the major problem for which one solution is to rely on a public key infrastructure (PKI) to vouch for end-node certificates is establishing trust between nodes. If end nodes have very limited computing resources, this may be infeasible.

In contrast, a blockchain is a distributed technology that guarantees the security of cryptocurrency networks. Nowadays, this innovative design has been widely used in many distributed applications [16–18], such as health care [19, 20] and VANET [21]. Various studies have also indicated that blockchain technology is suitable for solving the security problems of the IoT for the following three main reasons:

- *Decentralization* Blockchain technology possesses the same distributed topological structure as an IoT network, where the related data are decentralized and managed by all the agents involved in the network. Compared with conventional centralized management methods, distributed blockchain technology can avoid a single point of failure and greatly reduce the system maintenance cost for a large number of IoT devices.
- *No need for trust* Because of the energy-limited feature of IoT devices, it is a common phenomenon that trust among IoT devices is lacking. In this case, the data of devices can be leaked or compromised easily. Blockchain has the unusual characteristic that it cannot be tampered with, ensuring that entities can communicate with each other without trust. In the blockchain structure, the information is stored in the blocks with a unique hash sign, and each block possesses a hash backup of the previous block information. If unusual activity occurs, all members will be informed. This ubiquitous monitoring method can effectively avoid illegal tampering with the device data.
- *Anonymity* In most cases, the anonymity of the devices is also required. However, the configuration of most devices is relatively low and not suitable for traditional complex encryption methods. The address design of blockchain provides a new approach for IoT data security. Even if an adversary analyzes a series of

transactions that belong to the same address, she/he cannot obtain the device's real identity.

The arrival of the smart contract, called Blockchain 2.0, gives blockchain even more power, and increasingly more researchers have begun to focus on the property of custom permission control in IoT applications. The studies [22, 23] focus on access control problems. However, there have not yet been studies that address the design of a group key based on blockchain and smart contract characteristics. In this paper, we integrate the properties and advantages of blockchain into IoT communication and propose an efficient authenticated group key agreement protocol based on blockchain for IoT applications. However, it is not yet practical for most current resource-constrained IoT devices to interact directly with a blockchain.

For this reason, in the proposed protocol, we introduce a new entity called the device manager, who acts as an intermediary to connect IoT devices with blockchain networks and increases negotiation efficiency. Moreover, all consensus information between each device manager can be transmitted via transactions. In this design, the blockchain network can be treated as a secure consultation platform. According to our security analysis, the proposed protocol is secure against various attacks. Furthermore, the experimental results demonstrate that the proposed protocol has high negotiation efficiency.

The remainder of the paper is organized as follows. Related work for blockchain-based IoT applications is presented in Section 2. Section 3 illustrates the core components of the blockchain model used in the proposed protocol. The network architecture is presented in Section 4. Next, our design is proposed in Section 5. Then, we analyze the security and validity of the proposed protocol in Section 6. Section 7 makes some comparisons with traditional related protocols and provides the simulation results of the proposed protocol using Ethereum. Finally, Section 8 concludes this paper.

2 Related work

To verify the feasibility of blockchain-based IoT applications, an early study [24] utilized blockchain as a distributed service platform for IoT, and its performance analysis clearly indicates that distributed fog-node performance is significantly superior to that of a centralized cloud. Around the same time, Ali et al. [25] argued that the consensus mechanism of blockchain technology involves high bandwidth overhead and delays, meaning that it could not be directly applied to low-configuration IoT devices. Shortly afterward, a secure and lightweight architecture for smart-homes was described in detail [26]. Its design eliminates the overhead of blockchain while maintaining most privacy benefits. Subsequently, Kshetri [27] systematically evaluated blockchain's roles in strengthening security in the IoT and proposed several key underlying mechanisms related to blockchain-based IoT security. Then, Wang et al. [28] further introduced a reputable-based consensus protocol for blockchain-enabled IoT systems. Very recently, Bhushan et al. [29] provided a detailed survey on how to remodel blockchain to the unique IoT needs.

At present, the blockchain concept has been applied to many aspects of IoT technology. A bitcoin-based fair payment system for outsourcing the computation of fog devices was proposed [30]. Besides, as the concept of smart-cities develops, increasingly more IoT devices are connected to the Internet, creating a massive demand for storage space. Given the limitations of cloud storage, this will be a significant bottleneck in the development of IoT technologies. To address this problem, Shafagh et al. [31] proposed an auditable blockchain-based storage system to deal with this vast amount of data. Moreover, the key management of an intelligent transportation system and a wearable medical device control system based on blockchain technology have also been proposed [19, 32, 33].

3 Preliminaries

In this section, we briefly introduce the core component of the blockchain model used in the proposed protocol, including how to transfer the secret key negotiation information and store the identity information of the negotiation principal. At last, the consensus mechanism of blockchain is also described.

3.1 Transactions

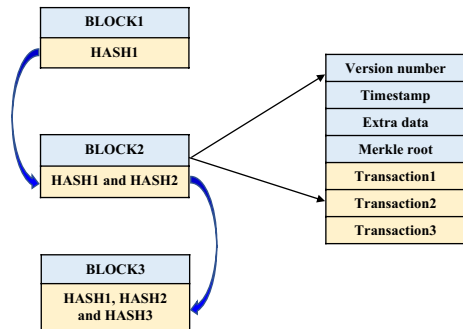
A transaction is the basic information unit that entities use to communicate with each other in the blockchain structure. In the first generation of the blockchain (i.e., the Bitcoin system), each of the information of a business trade of a client is regarded as a transaction, which mainly includes the identity of both parties and the amount of money exchanged. Namely, the object of a Bitcoin transaction can only be bitcoin, which severely reduces the range of blockchain applications. Ethereum, also called second-generation blockchain, removes this limitation and allows data of any type to be sent from an externally owned account to another account on the blockchain platform directly. Its primary strategy is to put all the data that needs to be transferred into one package and use the signature mechanism to guarantee its privacy.

A typical transaction in the Ethereum system consists of $\{from, to, value, data\}$ and other fields related to mining. The *from* field is filled with the sender's address; the *to* field is the goal address to which the information is sent, and the *value* field represents the value of the transmitted digital currency. Finally, the *data* field is used to store attachment information for the transaction and is always empty. In our proposed protocol, we put the required information for the authenticated group key generation process into the *data* field to make full use of this unused field. To implement this idea, we integrate all the negotiation information into a whole structure. As shown in Table 1, in addition to the basic information, the negotiation information includes the following main parts:

- *senderInfo* this field represents the detailed identity information of the originator who transmits the message.

Table 1 Information contained in the *data* field

SenderInfo	ReceiverInfo	Versions	Message	Type	Addition
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Fig. 1 Structure of the blockchain in our protocol

- *receiverInfo* this field represents the detailed identity of the goal information receiver.
- *versions* this field represents the time of negotiation.
- *message* this field represents devices' or device managers' personal opinions on the final group key.
- *type* this field is used to distinguish the generate process from the spread process.
- *addition* the function of this field is the same as that of *data*, which is used to store notes.

Note that in the proposed protocol, the blockchain model is used only to transmit negotiation information when generating the group key, and no transfer of currency is involved. Therefore, the *value* field should be set to zero.

3.2 Blockchain

As the name suggests, a blockchain is a chain of blocks connected in series [22, 23]. Each block, which has a unique hash value, contains a certain amount of transaction information, is the structure for storing negotiation information. These blocks are connected in chronological order by including a unique hash value for the previous block, ensuring the whole blockchain platform's security.

In the proposed protocol, the information flow also relies on the unique chain structure of a blockchain. Figure 1 shows the structure of the blockchain utilized in the proposed protocol, which is similar to that of Bitcoin. However, the contents in the block are slightly different. Here, the *Version number* field is used to record the number of group key agreements. The *Timestamp* field contains the generation time of a block. The *Merkle root* [34] field, which is responsible for connecting the block head and block body, is the pillar of a block. In addition to these basic information fields in a block, we also store the personalized information of the device managers

in the *Extra data* field, which is the field in which Satoshi Nakamoto wrote a news-letter in the Bitcoin system.

3.3 Consensus mechanism

In a traditional centralized service, the cloud has the highest and unique processing rights over user data. If the system changes or becomes corrupted, only centralized updates from the cloud server are required. In contrast to centralized design, in a distributed structure, to enable the whole blockchain network node to maintain the same data and guarantee the fairness of each participant, all participants in the system must have a unified protocol, that is, a consensus mechanism. Many kinds of consensus mechanisms have been proposed, such as *PoW* [35], *PoS* [36], *PoA* [37], *PoET* [38], and *PoRX* [39].

The *PoW* (Proof of Work) and *PoS* (Proof of Stake) mechanisms are the two most popular consensus mechanisms for blockchain applications. *PoW* is a protocol designed primarily to prevent and deter cyberattacks on the network. It is necessary to solve exceptionally difficult math problems to write a new trustless transaction group in the blockchain. When the *miner* completes this task, it obtains some rewards. The energy consumption required for this design is very high. Thus, the exorbitant cost of an attack often forces an attacker to abandon it.

Unlike the *PoW* system, in which the user validates transactions and creates new blocks by performing a certain amount of computational work, a *PoS* system requires the participant to show ownership of a certain number of cryptocurrency units. The creator of a new block is chosen in a deterministic way, depending on its wealth, which is also defined as a stake. A larger stake gives the user more book-keeping rights. Given the need for fast and efficient group key negotiations and the devices' limited energy, we deploy the *PoS* mechanism to achieve consistency in our work.

4 Network model

The proposed protocol consists of three participants: IoT devices, device managers, and the blockchain network. Figure 2 shows its architecture in detail.

- *IoT devices* IoT devices are a set of smart products (such as mobile phones, laptops, and smart-TVs) that we encounter within our daily lives. These energy-limited devices collect or send data for the network layer (e.g., the device manager or blockchain network). In the proposed protocol, all these IoT devices are divided into various groups, and each group is handled or managed by a device manager. Each IoT device must provide personal information to their device manager and search for a unique backup.
- *Device manager* As described above, the proposed protocol introduces a new role, the device manager, as an intermediary to connect IoT devices to a blockchain network. Each device manager is responsible for the centralized and uni-

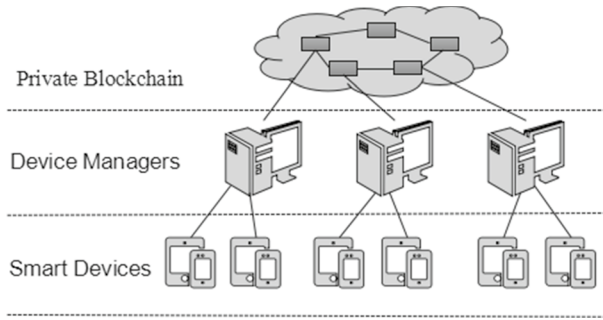


Fig. 2 Architecture of the proposed protocol

fied management of a subgroup of IoT devices and has the right to vote on a final group key. Besides, a device manager assigns a serial number to each IoT device it owns.

- **Blockchain network** In our design, we adopt a private (permissioned) blockchain model to maintain the distributed nature of the protocol. Unlike the public chain (permissionless), the private chain is only open to individuals or single entities, and the access and use of the data are under strict permission management. This architecture has extremely fast transaction speeds, better privacy, and lower transaction costs and is less vulnerable to malicious attacks. Considering the clustering characteristics of regional IoT devices, the use of a private blockchain can better address low negotiation efficiency.

All device managers are treated as a “strong authentication node” in our blockchain network. If a new device manager desires to join the private blockchain, the existing strong authentication nodes should approve this new device manager. The specific procedures for this approval are described as follows.

1. The new device manager sends its personal information to the blockchain network. Once the current conditions of that new device manager meet the need of our blockchain network, such as the validity of the timestamps and proof of adequate gas,¹ the new device node is permitted to join the blockchain network. Of course, this new device manager will also be treated as a “strong authentication node”.
2. The network prepares a public/private key pair for further use. Then, the network publicly publishes the public key and securely transmits the private key to the new device manager.

¹ To prevent malicious users from sending a large number of meaningless transactions in the blockchain network, resulting in the waste of computing resources, most of the existing blockchain platforms adopt the transaction charging strategy. For example, each transaction on Ethereum requires a fee, and gas is the basic unit used for measuring the computing resources consumed by a transaction. Generally, the more computational steps a transaction takes, the more gas it consumes.

Table 2 Notations used in the protocol

Notation	Description
g	A primitive element of multiplicative group $Z *_p$
p	A large prime number
n_A	Random number generated by the entity A (device manager)
P_A/R_A	Public key/private key of the entity A
$E_{P_A}(M)$	Encrypt M with P_A
$S_{R_A}(M)$	Sign M with R_A
T_i	The i^{th} transaction
mod	Modular operation
$\parallel$	Concatenation operation
$\oplus$	Bit-wise XOR operation
$H(\cdot)$	Hash function

3. After the network access is approved, the device manager applies for accounts for each device it owns.

5 Proposed protocol

In this section, we first propose a blockchain-based authenticated group key agreement protocol for IoT networks. We also discuss how to update a group key while adding/deleting a device manager.

5.1 Generation of the group key

Here, we discuss how to establish a group key based on blockchain technology. Notations used in the protocol are listed in Table 2. To make the description easy to understand, we assume three device managers named Alice, Bob, and Carol in the whole network. The overall consultation process is divided into the following five stages.

Issuing request When it is necessary to negotiate a secure group key GK , Alice, Bob, and Carol are negotiated within a specific order (e.g., according to the order in which they entered the group or their value to the group). In the proposed protocol, we initiate the consultation according to how the device managers joined the group. Assume that the order of these three device managers is Alice, Bob, and Carol. Under these circumstances, Alice is the first one to perform the procedure. Alice then performs the following steps.

- Step 1: Alice generates a random number n_{Alice} as her secret personal opinion and chooses the number g and p that are agreed by the consultative body (Alice, Bob, and Carol) in advance. Then, Alice calculates $N_{Alice} = g^{n_{Alice}} \bmod p$ for the group key and invokes other information $\delta_{Alice} = \{Alice, Bob, versions = 1, type = generate, addition = "null"\}$ in a transaction.

Step 2: Alice encrypts $(N_{Alice} \parallel \delta_{Alice})$ with Bob's public key P_{Bob} and signs $H(N_{Alice} \parallel \delta_{Alice})$ with Alice's private key R_{Alice} .

Step 3: Alice broadcasts the transaction information T_{Alice} to the whole blockchain network, where $T_{Alice} = E_{P_{Bob}}(N_{Alice} \parallel \delta_{Alice}) \parallel S_{R_{Alice}}(H(N_{Alice} \parallel \delta_{Alice})) \parallel timestamp$.

It should be noted that since g is primitive element of multiplicative group Z_p^* and p is a large prime number, considering the normal distribution characteristics of mod operation, n_{Alice} cannot be derived from $N_{Alice} = g^{n_{Alice}} \bmod p$ in reverse.

State delivery While Alice broadcasts the transaction T_{Alice} to the whole blockchain network, the default proxy node of the whole peer-to-peer blockchain platform (which also called miners) will act as qualified media workers and verify the validity of the T_{Alice} transaction that Alice has just broadcast. The default proxy node in the blockchain network does not execute transactions but it may verify them [40].

This pre-verification process mainly includes whether the timestamp is valid, whether the data format of the transaction is correct, and whether the token of the transaction is sufficient and other formal validation. If all the conditions are met, the proxy node push the transaction to the txpool² (transaction pool) and process all transaction in the txpool in order. Meanwhile, Alice is sent lock $H(T_{Alice})$, which uniquely identifies the communication message. In general, the processing speed of a transaction is proportional to the processing cost provided at the beginning of the transaction. A larger cost increases the speed of the transaction.

Identity authentication In the blockchain network, communication information is transmitted directly to the target communicator without the need for a third-party intermediary. After receiving T_{Alice} , Bob must verify the validity of T_{Alice} in the following steps.

Step 1: Bob decrypts $E_{P_{Bob}}(N_{Alice} \parallel \delta_{Alice})$ with his private key R_{Bob} and further obtains N_{Alice} .

Step 2: Bob can check the integrity of N_{Alice} by verifying the digital signature $S_{R_{Alice}}(H(N_{Alice} \parallel \delta_{Alice}))$ with Alice's public key P_{Alice} .

Message handling After checking the validity of the transaction he has received, Bob then performs the following steps.

Step 1: Bob selects a random number n_{Bob} as his own secret personal opinion and calculate $N_{Bob} = g^{n_{Bob}} \bmod p$ as Alice does. Then, he generates $\delta_{Bob} = \{Bob, Carol, versions = 1, type = generate, addition = "null"\}$ in a transaction.

Step 2: Bob encrypts $((N_{Alice} \oplus N_{Bob}) \parallel \delta_{Bob})$ with Carol's public key P_{Carol} and signs $H((N_{Alice} \oplus N_{Bob}) \parallel \delta_{Bob})$ using Bob's private key R_{Bob} .

² Txpool is a buffer for a large number of transactions. As long as there are new transactions, no matter they are created by this node or broadcast by other peer nodes, they will be first added to txpool. When the blocks are packed, transaction will be extracted from this pool.

Step 3: Bob broadcasts the transaction information T_{Bob} to the whole blockchain network, where $T_{Bob} = E_{P_{Carol}}((N_{Alice} \oplus N_{Bob}) \parallel \delta_{Bob}) \parallel S_{R_{Bob}}(H((N_{Alice} \oplus N_{Bob}) \parallel \delta_{Bob})) \parallel \text{timestamp}$.

In addition, proxy node (miners) involved in the whole blockchain network act as qualified media workers and verify the validity of T_{Bob} . If all the conditions are met, the *miners* push the transaction to the txpool. Eventually, Carol receives transaction T_{Bob} and checks its validity. After that, Carol can calculate final group key GK , where $GK = N_{Alice} \oplus N_{Bob} \oplus N_{Carol}$. Different from the centralized key distribution center, this group of keys contains the opinion information of all device administrators and can effectively avoid single point attack.

Consequence traversing Clearly, because it is a group key for the whole group, we should inform all other device managers, i.e., Alice and Bob, of outcome GK . Here, we follow the principle that the device manager who negotiates first will achieve the result first. Thus, when the final group key GK is determined, Carol first sends the final result to device manager Alice because she was the first device manager to initiate negotiations.

The process of group key negotiation is similar to the process described above.

Step 1: Carol encrypts $(GK \parallel \delta_{Carol})$ with Alice's public key and signs $(GK \parallel \delta_{Carol})$ using Carol's private key. Carol then transmits the transaction $E_{P_{Alice}}(GK \parallel \delta_{Carol}) \parallel S_{R_{Carol}}(GK \parallel \delta_{Carol}) \parallel \text{timestamp}$ to Alice. Note that $\delta_{Carol} = \{Carol, Alice, versions = 1, type = spread, addition = "null"\}$.

Step 2: While receiving this transaction, Alice checks the validity of this transaction and then generates another transaction $E_{P_{Bob}}(GK \parallel \delta_{Alice}) \parallel S_{R_{Alice}}(GK \parallel \delta_{Alice}) \parallel \text{timestamp}$ sends it to Bob. Also, $\delta'_{Alice} = \{Alice, Bob, versions = 1, type = spread, addition = "null"\}$.

Step 3: Bob now performs the same operation and sends a new transaction to Carol. At this time, Carol calculates GK again to ensure the information has not been modified. Under this premise, all device managers have established a common group key.

As for the distribution of the acquired group secret keys to the smart devices managed below, *MQTT* protocol can be run on low-end and battery-operated sensor/actuator devices and operate over bandwidth-constraint WSNs such as ZigBee-based networks and maybe an efficient methods to complete this communication mission. Since this part is not closely related to protocol design, we will not make further discussion. Those who are interested in this topic can refer literature [41].

5.2 Generalized to n device managers

Actually, as long as we follow a certain order, our agreement is not limited to three device managers. Assume there are n device managers, $M_1, M_2, \dots, M_n$, within the whole network. After $n - 1$ iterations, the last device manager M_n will receive the transaction $T_{M_{n-1}}$ in which

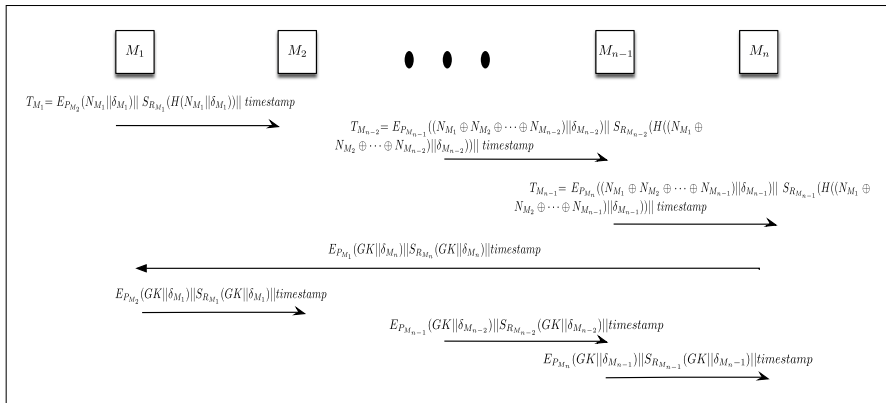


Fig. 3 Generalized to n device managers

$T_{M_{n-1}} = E_{P_{M_{n-1}}}((N_{M_1} \oplus N_{M_2} \oplus \dots \oplus N_{M_{n-1}}) \parallel \delta_{M_{n-1}}) \parallel S_{R_{M_{n-1}}}(H((N_{M_1} \oplus N_{M_2} \oplus \dots \oplus N_{M_{n-1}}) \parallel \delta_{M_{n-1}})) \parallel \text{timestamp}$. Device manager M_n now can calculate the group key GK , where GK is equal to $N_1 \oplus N_2 \oplus \dots \oplus N_n$. This group key contains the opinions of all legitimate device managers on the key and fully complies with the aforementioned fairness principle. Figure 3 illustrates the whole negotiation process.

5.3 Update of the group key

In fact, in the complex environment of the Internet of things, it is pervasive for a single intelligent device to join and leave. Since having classified the devices into subgroups, to minimize the energy consumption load of blockchain, we assigned the task of devices to update to the corresponding subgroups rather than do the re-keying procedure directly when a single device joins or leaves. As to the centralized management of devices for subgroups, there are many traditional protocols that can be used [42, 43]. Here we do not discuss the procedure.

On the contrary, although the update frequency of the device administrator with higher authority is very low, improper handling can threaten the security of the whole group communication. Therefore, in this part, we mainly discuss the re-keying procedure when the device administrator is leaving or joining.

Join operation Assume that a new device manager M_{n+1} desires to join the network. The existing group key should be updated to ensure the backward security of the group information [44]. First, M_{n+1} should register his/her account to the blockchain, and existing device administrators with representation rights update the local information list. After that, the $n + 1$ device managers can negotiate a new group key by performing the steps mentioned above.

Leaving operation The departure of the device manager is generally divided into active and passive leave two kinds of circumstances [45]. The so-called passive leave means that the participant is not safe or has faults and needs to be replaced. In another case, the device administrator applies to leave voluntarily.

Table 3 Explanation of basic symbols in BAN logic system

Notation	Description
A, B	Specific communication subject
K_a, K_b	Public key of specific communication subject
K_a^{-1}, K_b^{-a}	Secret key of specific communication subject
(X, Y)	Connection of X and Y
$A \models X$	A believes X
$A \triangleleft X$	A sees X and received a message containing X
$A \mid \sim X$	A once said X
$A \mid \Longrightarrow X$	A has jurisdiction or control over X
$\sharp(X)$	X is fresh
$A \xleftrightarrow{K} B$	A and B use the shared key K to communicate
$A \stackrel{X}{\rightleftharpoons} B$	A and B share secret X , other subjects do not know X
$\{X\}_K$	X is encrypted by key K
$\langle Y \rangle_X$	Y contains secret X

In the passive leave, we only need to distribute the managed devices to other administrators for secondary management and do not need to update the group secret keys. However, when an active leave happens, to prevent one or more device managers who have left from using the previous group key, the group key should be updated. The re-keying procedure as follows:

- Step 1: The leaving device manager M_i trigger remaining $n - 1$ device managers' smart contract *Changing*, where calculate the reverse value N'_i of $N_i = g^{n_i} \bmod p$ and can inform other device manager that remove the address information of the leave.
- Step 2: Leaving device manager M_i performs a self-destruct operation *Destruction* to update the device list, thus ensuring that non-listed devices cannot access the information in the remaining group.
- Step 3: The remaining device manager calculates the new group key GK' in local by multiplying the reverse value N'_i of the leaver, where $GK' = N_1 \oplus N_2 \oplus N_{i-1} \oplus N_{i+1} \cdots \oplus N_N$.

6 Security analysis

Here we demonstrate that the proposed protocol is indeed secure with BAN logic. BAN logic was proposed by Burrows et al. [46] and is a modal logic based on belief and knowledge. It has become the most famous tools and widely used to analyze the security of authenticated key agreement protocols† [47].

6.1 Definition and rules of BAN logic

To facilitate the subsequent use of BAN logic to analyze the security of the protocol, Table 3 briefly introduces some basic symbols and communication rules used in the BAN logic system.

The communication rules are a form of representation like $\frac{P}{Q}$. The basic meaning of the expression is that if P is correct, then Q is also correct. The BAN logic has detailed regulations on message meaning, message freshness, message reception, and jurisdiction.

Message meaning rule (R1) If A believes that the shared key between A and B is K , and A has received the message $\{X\}_K$ encrypted with K , then A believes that B has said X information:

$$\frac{A \models A \xleftrightarrow{K} B, A \triangleleft \{X\}_K}{A \models B \sim X} \quad (1)$$

(R2) If A believes that K is a shared secret between A and B , and A has received a message X containing Y , then A believes that B once said X :

$$\frac{A \models A \xleftrightarrow{Y} B, A \triangleleft \langle X \rangle_Y}{A \models B \sim X} \quad (2)$$

The nonce-verification rule (R3) If A believes that X is fresh, and A believes that B has sent X , then A believes X :

$$\frac{A \models \sharp(X), A \models B \sim X}{A \models B \equiv X} \quad (3)$$

Jurisdiction rule (R4) If A believes that B has the right to control X , and A believes that B also believes in X , then A believes in X :

$$\frac{A \models B \implies X, A \models B \equiv X}{A \models X} \quad (4)$$

The freshness rule (R5) If A believes that X is fresh, then A believes that the overall cascading information of X and Y is also fresh.

$$\frac{A \models \sharp(X)}{A \models \sharp(X, Y)} \quad (5)$$

Receive message rule (R6) If A has ever received the message (X, Y) , then A has received the component X of the received message.

$$\frac{A \triangleleft (X, Y)}{A \triangleleft X} \quad (6)$$

(R7) If A has ever received a message $(X)_Y$ containing secret Y , then A must have received X .

$$\frac{A \triangleleft (X)_Y}{A \triangleleft X} \quad (7)$$

The belief rule (R8) If A believes in X and A also believes in Y , then A believes in the message (X, Y) . If A believes in the message (X, Y) , then A also believes in X and Y .

$$\frac{A \models X, A \models Y}{A \models (X, Y)} \quad (8)$$

$$\frac{A \models (X, Y)}{A \models X} \quad (9)$$

(R9) If A believes B believes (X, Y) , then A believes B believes X and Y .

$$\frac{A \models B \models (X, Y)}{A \models B \models X} \quad (10)$$

6.2 BAN logic formalized security proof

This section mainly uses the formal analysis method (BAN logic) to analyze the security of the designed key agreement protocol. To simplify the proof process, we limit the number of negotiation entities to two. Member u_1 is the first negotiation entity, and member u_2 represents the last negotiation entity. In a distributed negotiation environment, the negotiation method is fixed, so the effect of specifying two negotiation entities for logical proof is the same as the effect of specifying n negotiation entities. Using BAN logic to analyze the security of a protocol is divided into the following four steps: idealizing protocol messages, formalizing protocol goals, initializing assumptions, and specific reasoning processes.

6.2.1 Idealizing protocol messages

BAN logic has its own unique set of logical symbols. Therefore, before using formal logic for protocol security analysis, the protocol in Fig. 3 needs to be described using symbols unique to BAN logic.

$$\begin{aligned} \text{Msg1} \quad & u_2 \triangleleft (a, y_1, t_1, u_1 \xleftrightarrow{GK} u_2)_{pub} \\ \text{Msg2} \quad & u_1 \triangleleft (b, y_2, t_2, u_1 \xleftrightarrow{GK} u_2)_{pub} \end{aligned}$$

6.2.2 Formalizing protocol goals

The design purpose of the key agreement protocol is to enable each negotiation entity involved in the negotiation to establish a group key used only for internal members, so as to use the group key to ensure the security of subsequent communications. Therefore, the group key agreement protocol designed based on blockchain

technology in this paper has the following four security goals: G1–G4. Targets G1 and G2 indicate that u_1 and u_2 believe that they have established a shared GK with each other. The targets G3 and G4 mean that u_1 and u_2 believe that the other party already knows the key GK for communication between the two.

$$G1 \ u_1 \models u_1 \overset{GK}{\longleftrightarrow} u_2.$$

$$G2 \ u_2 \models u_1 \overset{GK}{\longleftrightarrow} u_2.$$

$$G3 \ u_1 \models u_2 \models u_1 \overset{GK}{\longleftrightarrow} u_2.$$

$$G4 \ u_2 \models u_1 \models u_1 \overset{GK}{\longleftrightarrow} u_2.$$

6.2.3 Initializing assumptions

Further extracting hypothesis descriptions from our agreement will help our entire logical proof process to be clearer. According to the key agreement protocol in the local IoT environment we designed, the following assumptions can be abstracted, where pub represents the secret information shared between u_1 and u_2 . SK represents the shared key information between u_1 and u_2 .

$$A1 \ u_1 \models u_1 \overset{pub}{\rightleftharpoons} u_2.$$

$$A2 \ u_2 \models u_1 \overset{pub}{\rightleftharpoons} u_2.$$

$$A3 \ u_1 \models \#(t_N).$$

$$A4 \ u_2 \models \#(t_N).$$

$$A5 \ u_1 \models u_2 \implies u_1 \overset{SK}{\longleftrightarrow} u_2.$$

$$A6 \ u_2 \models u_2 \implies u_1 \overset{SK}{\longleftrightarrow} u_2.$$

6.2.4 Detailed proof process of the protocol

Through the use of formal messages, abstract assumptions and logical inference rules, the goals G1–G4 of the protocol can be finally obtained.

By Msg1, A2 and message meaning rule R2, we can infer

$$\mathbf{P1:} \ u_2 \models u_1 \mid \sim (a, y_1, t_1, u_1 \overset{GK}{\longleftrightarrow} u_2).$$

From A4 and the freshness rule R5, the hypothetical sentence P2 can be obtained:

$$\mathbf{P2:} \ u_2 \models \#(a, y_1, t_1, u_1 \overset{GK}{\longleftrightarrow} u_2).$$

Using P1, P2, and R3, the statement P3 can be inferred:

$$\mathbf{P3:} \ u_2 \models u_1 \models (a, y_1, t_1, u_1 \overset{GK}{\longleftrightarrow} u_2).$$

According to P3 and R8, we get

$$\mathbf{P4:} \ u_2 \models u_1 \models u_1 \overset{GK}{\longleftrightarrow} u_2. \ (\mathbf{G4})$$

The statement P5 can be inferred from P4, A6, and R4:

$$\mathbf{P5:} \ u_2 \models u_1 \overset{GK}{\longleftrightarrow} u_2. \ (\mathbf{G2})$$

Table 4 Formula for key generation overhead

Protocol	Cost	Key number	Central-ized server
Simple app	n^2	1	Y
LKH [48]	$n * [\log_2 n]$	1	Y
Alireza et al. [45]	$n * \log_2 [n/m]$	1	Y
DBGK [49]	$[n/m] * [\log_2 m]$	n/m	Y
Ours	$2 * [n/m]$	1	N

Similarly, according to the message Msg2, assumption A1 and rule R2, we get

P6: $u_1 \models u_2 \mid \sim (b, y_2, t_2, u_1 \xleftrightarrow{GK} u_2)$.

Using A3 and R5, we get

P7: $u_1 \models \#(b, y_2, t_2, u_1 \xleftrightarrow{GK} u_2)$.

Using P6, P7, and R3, we derive

P8: $u_1 \models u_2 \models (b, y_2, t_2, u_1 \xleftrightarrow{GK} u_2)$.

By P8 and R8, we get

P9: $u_1 \models u_2 \models u_1 \xleftrightarrow{GK} u_2$. (G3)

According to P9, A5, and R4, we can infer

P10: $u_1 \models u_1 \xleftrightarrow{GK} u_2$. (G1)

From the above analysis, it can be seen that the goals G1, G2, G3, and G4 of the final protocol can all be derived, that is, the key agreement protocol designed based on the blockchain technology in this paper can achieve the expected security goals.

7 Comparison and performance evaluation

In this section, we first compare it with other similar protocols. We then examine the feasibility and performance of blockchain-based group key agreement protocol by simulating a permissioned blockchain in the Ethereum. The performance evaluation is broken into three parts. First, we give the simulation hypothesis in the start part. Next, we study the time cost in each phrase of our proposed protocol. At last, we simulate the protocol scalability by adding/removing device administrators.

7.1 Comparison of blockchain and traditional structures

To evaluate the performance of our blockchain-based protocol, we compare the cost of sending notifications to the involved device managers with other traditional distributed group key protocols: simpleApp, LKH [48], DBGK [49], and Alireza et al. protocol [45]. The compare result, as shown in Table 4, where n represents the total number of smart devices and m represents the number of devices in each subgroup.

From the above data, it can be seen that the communication load of SimpleApp is proportional to the number of devices n . In contrast, the number of devices on the Internet of things environment is vast; thus, this method is often taken as the worst

Table 5 Information of other admitted nodes

No.	Operating system	CPU	RAM (GB)	Disk (GB)	LAN IP
1	Ubuntu 16.04	Intel(R) i5-7400 CPU @ 3.40 GHZ	7.7	25.0	219.223.246.88
2	Ubuntu 16.04	Intel(R) i5-7400 CPU @ 3.40 GHZ	7.7	25.0	219.223.243.29
3	Ubuntu 16.04	Intel(R) i5-7400 CPU @ 3.40 GHZ	7.7	25.0	219.223.240.225

comparison standard and is rarely used. LKH introduces the logical key tree concept and reduces a load of secret key negotiation from linear length n to binary tree height $\log_2 n$.

However, when the number of devices is large, the traditional binary logic tree's height will also be high, significantly reducing the efficiency of secret key negotiation. Alireza et al. [45] proposed a novel group key management protocol that extends the number of devices in each subgroup from 2 to m , when m is large, the protocol has proved to be much more efficient than traditional methods LKH. However, both the traditional LKH method and the improved LKH method [45] adopt the centralized design mode, and each secret key update requires the centralized secret key server to update from scratch, which is easy to be attacked by a single point. In the DBGK [49], each subgroup is assigned a different group key to realizing the local update of the secret key. However, although the local update of the secret keys is fast, multi-secret keys bring new obstacles to the communication between subgroups and increase the delay on the whole. Unlike the centralized design, we will negotiate the principal agreement between n device managers, and each device manager needs only two times of communication. In conclusion, compared with other protocols, our blockchain-based protocol is more secure and efficient.

7.2 Simulation hypothesis

In the proposed protocol, we represent each device manager with a node on the blockchain and manage the underlying devices through blockchain accounts. First, we build a permissioned Ethereum network under certain circumstances (e.g., the blockchain is owned by a specific organization and only the devices under the same local area network (LAN) are allowed to be added), which is executed on a desktop with an Intel (R) Core (TM) i5-3470 CPU @ 3.20 GHZ and 7.7 GB RAM running Ubuntu 16.04. The programming language used in this implementation is Go. The difficulty of each block is set to 0x0200 and the gas limit cost is set to 0x2fe8d8, which be used to maintain the efficiency of negotiation. This security level is enough to secure the procedure because device managers M_i are trusted participant [32]. Permissioned networks often require less traffic load than permissionless networks. Therefore, it is more in line with the energy needs of IoT devices.

To guarantee the privacy of the permissioned chain, we confine other device managers to the same domain and access to the network is strictly based on the access rules we specified above. Only authorized administrators can access and use

Table 6 Time cost for initialization operations

Phase	Time (ms)
Identity registration	3.512
Comparison policy	2.583
Account allocation	1.436
Device list generation	4.379
Device binding	5.448

Table 7 Operations performed during the generation of group key

Phase	Operations
Issuing request	NOrder.Sort + Info.Setout + Info.Enc + Info.Sign
State delivery	Block mining
Identity authentication	TStamp.Ver + RInfo.Dec + RInfo.Ver
Message handling	AInfo.Merg + NInfo.Setout + NInfo.Enc + NInfo.Sign
Consequence traversing	TOrder.Sort + GK.Enc + GK.Sign + TStamp.Ver

Table 8 Average processing time for main schedules

Processing schedules	Time cost (ms)
Order sorting	0.115
Modular exponentiation	5.524
Verify timestamp	0.101
ECC encryption	0.539
ECC decryption	0.743
ECDSA signing	0.553
ECDSA verifying	1.132
Block mining	4.174

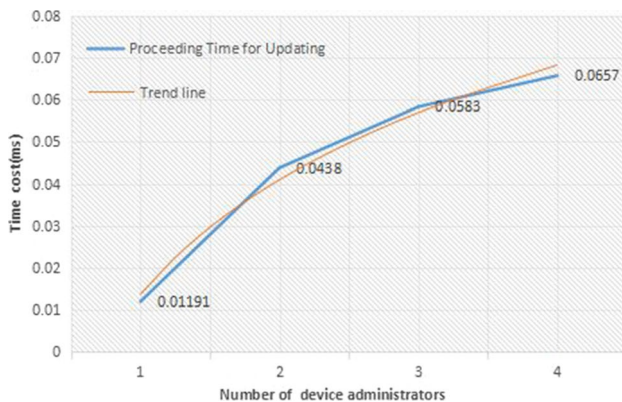
the device information in the whole network. The detailed settings for the simulation are shown in Table 5.

7.3 Performance evaluation of the consultation process

A good platform is a prerequisite for secret key negotiation. Thus, at first, we study the processing time for initialization. Identity registration is the first step for each device manager to gain authorization. Each device manager M_i invokes a series of personal information registrations and management measures for storage in a unique block, and the entire registration process is subject to the prerequisites policy. Then, permissioned device manager assigns an individual account to each physical device they manage and number them. At last, after a series of processes, each device manager obtains a list of devices that contains accounts and

Table 9 Execution time of the proposed protocol

Phase	Time (ms)
System initialization	17.358
Issuing request	6.731
State delivery	4.174
Identity authentication	1.976
Message handling	6.938
Consequence traversing	6.717

**Fig. 4** Variation of the number of device managers

numbers. The resulting average time cost of each operation in the initialization is presented in Table 6.

What is more, we use two device managers to perform a test on the proposed protocol and evaluate the time cost of each phase in the generation of a group key. First, we summarize the operations (e.g., the determination of the deliberative order, the formation of the information lists, and the series of encryption and decryption algorithms used in an identity verification) during the group key consultation in Table 7 and records the average processing time for each main operation in Table 8.

To better verify the validity of the proposed protocol, we summarize the time consumed for the whole group key negotiation process in Table 9. The most time-consuming operations occur in the initialization phase. This is because the device managers need to build a permissioned blockchain by themselves. However, the total time cost of such blockchain-based group key agreement protocol is nearly 0.73s, which is half of the time that be used to hand up the traditional structure.

7.4 Simulation of the dynamics of group membership

Although blockchain technology provides adequate security for many applications, its high latency also has been criticized. Any efficient group key negotiation protocol

cannot ignore a dynamic change in-group members. Thus, in this part, we perform a simulation experiment to evaluate the proposed protocol dynamics when adding two device managers in turn. The configuration information of the two device managers is shown in Table 5. Note that, to simulate IoT devices' clustering characteristics and achieve the conditional access of each physical device to the entire IoT environment, we still restrict each device manager to the same LAN. An example of the energy consumption required for an increase in equipment using our proposed protocol is shown in Fig. 4.

The simulation results show that the time consumed for group key negotiation is linearly related to the group members' size. When the number of members increases, the time required for a group key negotiation does not increase substantially. Therefore, we can make the bold prediction that the protocol is entirely suitable for the group key negotiation of a cluster of IoT devices within a specific range, no matter how extensive the area of the cluster. Additionally, the group key is not changed frequently. Thus, the storage space required for determining this information is also minimal, which resolves the issues of the energy limitations of IoT devices and mitigates their communication costs.

8 Conclusion

The distributed and security features of the blockchain provide a new solution to group key negotiation for IoT applications. First, we described how we relay communications on the blockchain network to generate a public–private key pair for the devices to avoid a traditional attack on a centralized server. Then, we outlined the core components of the local blockchain in our system and discussed the process of group key establishment and updating concerning the three scenarios of initialization, adding a member, and removing a member. We also presented a comprehensive security and privacy analysis of the proposed protocol. Our simulation results demonstrate that the overheads involved in our method are extremely low, which fully meets the energy and information processing limitations of IoT devices.

In the future, we will focus our research on the secret key negotiation of IoT devices in different domains and implement access control of the group information with the more complicated intelligent contract mechanism of the Ethereum platform, aiming to increase the scalability of the proposed protocol.

Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflict of interest.

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